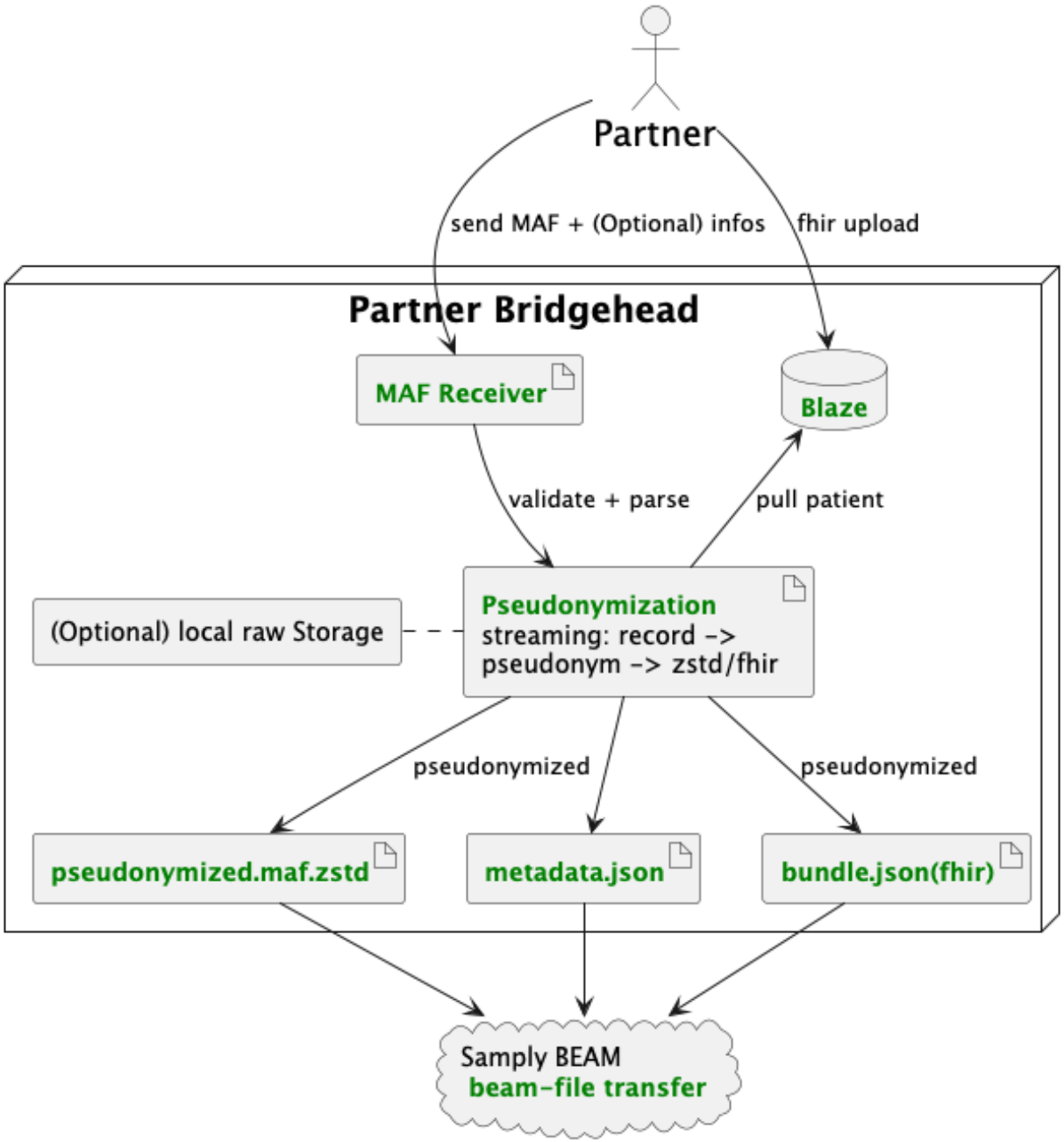
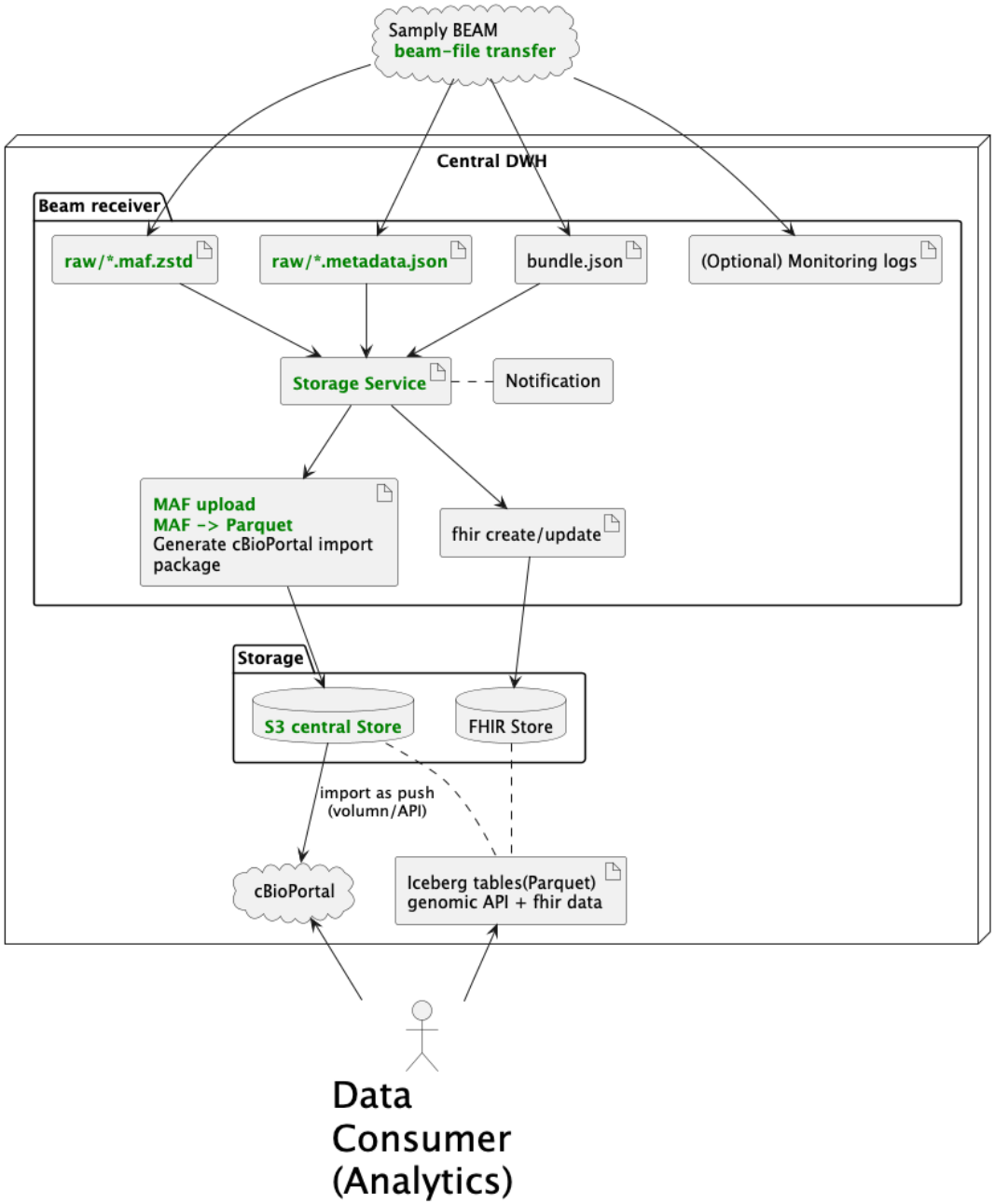
March 16, 2026

Proposed Architecture Ingest Bridgehead and Central DWH Version 1

ITCC DHW


Partner
send MAF + (Optional) infos
fhir upload
Partner Bridgehead
MAF Receiver
Blaze
validate + parse
pull patient
(Optional) local raw Storage
Pseudonymization
streaming: record -> pseudonym -> zstd/fhir
pseudonymized
pseudonymized
pseudonymized.maf.zstd
metadata.json
bundle.json(fhir)
Samply BEAM
beam-file transfer


Samply BEAM
beam-file transfer
Central DWH
Beam receiver
raw/*.maf.zstd
raw/*.metadata.json
bundle.json
(Optional) Monitoring logs
Storage Service
Notification
MAF upload
MAF -> Parquet
Generate cBioPortal import package
fhir create/update
Storage
S3 central Store
FHIR Store
import as push
(volumn/API)
cBioPortal
Iceberg tables(Parquet)
genomic API + fhir data
Data Consumer (Analytics)

Endpoints Ingest

Required Patient ID

- Fields to extract and pseudomize patient id `Tumor_Sample_Barcode` and `Matched_Norm_Sample_Barcode`
- Expected format: `prefix{patientID}_suffix(custom)`

1. **POST /omics/upload**
   - Upload MAF files (binary body)
   - Max body size: 0.5 GB
   - Requirement patient data (FHIR) need to be provided
   - Validates and scans input before processing
   - Pseudonymisation before sending to Central DWH
2. **GET /omics/health**
   - Health check endpoint
   - Returns: `200 OK`
   - Response body: `"OK"`

Required Fields in MAF File

Mandatory Columns (Header):

- Hugo_Symbol
- NCBI_Build
- Chromosome
- Start_Position
- End_Position
- Variant_Classification
- Variant_Type
- Reference_Allele
- Tumor_Seq_Allele1
- Tumor_Seq_Allele2
- Tumor_Sample_Barcode
- Matched_Norm_Sample_Barcode

cBioPortal – Minimal Generated Structure

Generated Study Directory:

```
partner_id/File_id(binary hash maf file)/
|- meta_study.txt
|- meta_clinical_sample.txt          -> data_clinical_sample.txt
|- meta_clinical_patient.txt         -> data_clinical_patient.txt
|- meta_mutation.txt                 -> data_mutation.maf
```

Notes:

- Generated automatically from FHIR data in Central DWH
- Each data file requires a corresponding meta file
- Meta files describe type, format, and processing details
- Data files contain the actual omics / clinical FHIR content